Comparative Privacy and Utility Analysis of l-Diversity Anonymization Algorithms

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Abstract: The increasing availability of microdata for data analytics raises significant concerns regarding sensitive attribute disclosure. Although k-anonymity reduces the risk of identity re-identification, it remains vulnerable to attribute inference attacks when sensitive values within an equivalence class are insufficiently diverse. To mitigate this limitation, l-diversity introduces additional constraints on the distribution of sensitive attributes. However, stronger privacy protection often comes at the expense of data utility. This study presents a comparative evaluation of three l-diversity models : Distinct, Recursive, and Entropy across four anonymization algorithms. Experiments are conducted on the Adult dataset and evaluated using information loss, classification accuracy, and disclosure risk. The results reveal a consistent privacy utility tradeoff among the evaluated models. Distinct l-diversity preserves data utility most effectively, achieving low information loss and stable predictive performance, but provides limited protection against attribute inference. In contrast, Entropy l-diversity offers the strongest privacy protection by substantially reducing disclosure risk, although it incurs considerable utility degradation at higher diversity levels. Recursive (c, l)-diversity provides a balanced compromise between privacy and utility, with $c = 3$ yielding the most favourable results across the evaluated metrics. Furthermore, the k-member repair algorithm, by implementing a swap and merge mechanism, demonstrates competitive performance by maintaining low information loss while effectively satisfying diverse l-diversity requirements. These findings highlight the importance of selecting an appropriate l-diversity model based on the desired balance between privacy protection and data utility in privacy-preserving data publishing.

Keywords: l-diversity, disclosure risk, sensitive attribute inference, data utility, extending k-anonymity to l-diversity, swap and merge mechanism.

1. Introduction

The increasing availability of data has created significant opportunities for research, analytics, and machine learning. However, publishing datasets that contain personal information also raises privacy concerns. Even after direct identifiers are removed, attackers may still infer sensitive information such as health conditions, income levels, or personal beliefs from the released data.

One of the main privacy threats is sensitive attribute inference, where an attacker uses quasi-identifiers and background knowledge to predict an individual's sensitive value. Traditional privacy models such as k-anonymity [1] reduce the risk of identity disclosure but do not adequately protect against attribute inference. An equivalence class may satisfy k-anonymity while still containing highly similar or dominant sensitive values, allowing sensitive information to be inferred with high confidence [2].

To address this limitation, l-diversity [2] was introduced to ensure sufficient diversity of sensitive values within each equivalence class. Several variants of l-diversity have been proposed, including Distinct l-diversity, Recursive (c, l)-diversity, and Entropy l-diversity [2]. Each provides different levels of protection against attribute disclosure. While stronger diversity requirements can reduce inference risk, they may also increase information loss and reduce data utility.

Therefore, understanding the tradeoff between privacy protection and data utility is essential for practical data publishing. Although l-diversity has been widely studied, the comparative impact of different l-diversity models on utility preservation remains insufficiently explored.

This study presents a comparative analysis of Distinct, Recursive, and Entropy l-diversity using multiple anonymization algorithms. The evaluation examines the effects of stronger l-diversity on information loss, classification accuracy, inference risk, and computational efficiency. By analyzing the privacy–utility tradeoff across different diversity models, this work provides insights into selecting an appropriate l-diversity model for protecting sensitive attributes while maintaining data utility.

The main contributions of this study are as follows:

(1) A comprehensive comparative evaluation of Distinct, Recursive, and Entropy l-diversity across anonymization algorithms, investigating their impact on informa-

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tion loss, data utility, and disclosure risk.

(2) Practical recommendations for selecting suitable privacy models based on application requirements.

The rest of this paper is organized as follows. Section 2 introduces the theoretical background of privacy-preserving data publishing and l-diversity as a defense against sensitive attribute inference. Section 3 presents the research methodology, anonymization algorithms, and evaluation metrics used for comparative analysis. Section 4 presents and discusses the experimental results, examining the impact of Distinct, Recursive, and Entropy l-diversity on privacy and utility. Finally, Section 5 summarizes the findings and suggests avenues for future work.

2. Background and related work

This section reviews the definitions of k-anonymity and l-diversity, followed by an overview of representative anonymization techniques evaluated in this study.

2.1 k-Anonymity

Let $D(A_1, A_2, \ldots, A_d, SA)$ denote a microdata table, where $QI = \{A_1, A_2, \ldots, A_d\}$ represents the set of quasi-identifier attributes and SA denotes a sensitive attribute. An equivalence class $EC \subseteq D$ is defined as a maximal group of records sharing identical values for all quasi-identifiers in QI.

Definition 1 (k-Anonymity): A dataset D satisfies k-anonymity if every equivalence class contains at least k records:

$$|EC_i| \geq k, \quad \forall EC_i \in D. \tag{1}$$

By ensuring that each record is indistinguishable from at least $k-1$ other records with respect to the quasi-identifiers, k-anonymity reduces the risk of identity disclosure. However, it does not adequately protect against attribute disclosure when sensitive values within an equivalence class are homogeneous or highly skewed.

2.2 l-Diversity

To mitigate attribute disclosure, l-diversity extends k-anonymity by requiring sufficient diversity of sensitive values within each equivalence class. Let $S(EC_i)$ denote the multiset of sensitive values in equivalence class EC_i, S' denote the domain of the sensitive attribute S, and $p(EC_i, s)$ denote the relative frequency of sensitive value $s \in S'$ within EC_i.

2.2.1 Distinct l-Diversity

Definition 2 (Distinct l-Diversity): An equivalence class EC_i satisfies Distinct l-diversity if it contains at least l distinct sensitive values:

$$|SA(EC_i)| \geq l. \tag{2}$$

Distinct l-diversity is simple to implement and generally preserves data utility. However, it does not consider the distribution of sensitive values and therefore remains vulnerable to probabilistic attribute disclosure when one sensitive value dominates the equivalence class.

2.2.2 Recursive (c, l)-Diversity

Distinct l-diversity may be insufficient when sensitive values exhibit highly skewed distributions. Recursive (c, l)-diversity addresses this limitation by restricting the dominance of the most frequent sensitive value.

Let $r_1 \geq r_2 \geq \cdots \geq r_m$ denote the frequencies of sensitive values in EC_i sorted in descending order.

Definition 3 (Recursive (c, l)-Diversity): An equivalence class satisfies Recursive (c, l)-diversity if

$$r_1 < c \sum_{j=l}^{m} r_j, \tag{3}$$

where $c > 0$ is a defined parameter controlling the strictness of the diversity requirement.

Recursive (c, l)-Diversity limits the dominance of the most frequent sensitive value and provides stronger protection against attribute disclosure than Distinct l-Diversity.

2.2.3 Entropy l-Diversity

Definition 4 (Entropy l-Diversity): An equivalence class EC_i satisfies Entropy l-Diversity if

$$H(EC_i) = -\sum_{s \in S'} p(EC_i, s) \log p(EC_i, s) \geq \log(l), \tag{4}$$

Entropy l-Diversity considers the entire distribution of sensitive values and requires sufficient uncertainty within each equivalence class. Consequently, it provides stronger protection against attribute disclosure than Distinct l-Diversity and Recursive (c, l)-Diversity.

2.3 Overview of l-Diversity Algorithms

Numerous anonymization algorithms have been proposed to satisfy l-diversity while preserving data utility. Existing approaches can generally be categorized into two groups. The first group generates k-anonymous and l-diverse datasets simultaneously, including Improved Scalable l-Diversity (ImSLD) [3] and UnitAnony [4]. The second group adopts a repair-based strategy that transforms an existing k-anonymous dataset into an l-diverse dataset, such as Systematic Clustering [5] and k-member l-diversity [6].

These algorithms differ in their anonymization strategies and consequently exhibit different levels of privacy protection, information loss, and computational efficiency. Therefore, a comparative evaluation is necessary to understand the trade-offs among alternative approaches and to identify suitable algorithms for privacy-preserving data publishing.

2.3.1 Improved Scalable l-diversity (ImSLD) algorithm

ImSLD is a scalable clustering-based anonymization approach for privacy-preserving data publishing. The algorithm orders quasi-identifiers according to domain size, constructs initial equivalence classes, and iteratively merges them until both k-anonymity and l-diversity constraints are satisfied. To minimize information loss, ImSLD applies local recoding by generalizing values only within affected equivalence classes rather than across the entire dataset. Mehta

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et al. [3] further enhanced this approach by ensuring that each equivalence class contains at least l distinct sensitive attribute values, thereby strengthening protection against attribute disclosure attacks. However, despite achieving relatively low information loss, ImSLD does not explicitly optimize information loss during equivalence class formation and instead assumes that sequentially ordered equivalence classes are sufficiently similar for merging, which improves scalability but may lead to suboptimal clustering and non-minimal information loss.

2.3.2 UnitAnony algorithm

UnitAnony [4] is an anonymization framework for privacy-preserving data publishing that combines record grouping, lattice-based generalization, and local recoding. The algorithm first groups records with identical QID values into units to reduce redundancy and improve efficiency. Next, the algorithm explores candidate anonymization schemes through lattice traversal, where each scheme represents a different generalization level for the QID attributes. Under each scheme, records are merged into equivalence classes (ECs) based on their transformed QID values. These ECs are then evaluated against k-anonymity and l-diversity constraint. ECs that satisfy the privacy constraints are retained, while ECs that fail to meet the requirements are further generalized in subsequent iterations. Once a valid anonymization scheme is identified, local recoding is applied within each EC. If no valid anonymization scheme can satisfy the constraints, the remaining records are suppressed.

2.3.3 Systematic clustering algorithm

Kabir et al. [5] proposed the systematic clustering method for l-diversity to address both identity and attribute disclosure risks. The algorithm first constructs k-anonymous clusters by sorting records based on their quasi-identifiers and initializing multiple clusters using selected seed records. Remaining records are then iteratively assigned to the clusters that produce the minimum increase in information loss. After the clustering stage, the algorithm evaluates the diversity of sensitive attribute values within each cluster. Clusters that do not satisfy the required l-diversity threshold are dissolved, and their records are reassigned to valid clusters using greedy assignment based on the minimum information loss criterion. This two-stage process ensures that the final anonymized dataset satisfies both k-anonymity and l-diversity, although the greedy assignment strategy may result in suboptimal cluster compositions and increased information loss.

2.3.4 *k*-member algorithm for *l*-diversity (swapping records & merging ECs)

The framework, presented in Algorithm 1, transforms a k-anonymous dataset into an l-diverse dataset through a repair process that begins by applying the original k-member algorithm to the dataset into ECs. Each generated EC is subsequently evaluated against the specified l-diversity constraint and categorized into two groups: valid equivalence classes (EC_{good}), which satisfy both k-anonymity and l-diversity, and violating equivalence classes (EC_{bad}), which require further repair.

Algorithm 1 Transforming k-anonymous data into l-diverse data through repairing ECs [6]

Input: k-anonymous equivalence classes EC, diversity parameter l
Output: l-diverse equivalence classes EC_{final}

```
Pool ← all records in EC;
EC_final ← ∅;
while Pool ≠ ∅ do
    Generate equivalence classes EC from Pool;
    Partition EC into valid classes EC_good and violating classes EC_bad;
    foreach violating equivalance class EC_v ∈ EC_bad do
        // Phase 1: Swap
        while EC_v violates l-diversity do
            Find nearby equivalence classes EC_s ∈ EC_good according to seed distance;
            Identify dominant sensitive values and missing sensitive values in EC_v;
            Generate feasible swap candidates;
            Select the candidate with minimum increasing information loss;
            if a feasible swap exists then
                Perform the swap;
            else
                Break;
        // Phase 2: Rebuild
        if EC_v still violates l-diversity then
            Reconstruct EC_v using records from unresolved violating classes;
        // Phase 3: Merge
        if EC_v still violates l-diversity then
            Find candidate equivalence classes for merging;
            Select the merge with minimum information loss cost;
            Perform the merge;
    Move all valid and repaired equivalence classes to EC_final;
    Pool ← records belonging to unresolved violating classes;
return EC_final;
```

For each violating equivalence class EC_v in EC_{bad}, the framework applies three repair phases: swap, rebuild, and merge. In the swap phase, the algorithm identifies the dominant sensitive value and the set of missing sensitive values required to satisfy l-diversity. Candidate donor equivalence classes EC_s are selected from EC_{good} and are ranked by the distance of the cluster seeds. Starting from the nearest donor EC_s, the algorithm searches for records containing the missing sensitive values and generates candidate swaps by exchanging a dominant-value record in EC_v with a donor record.

A swap is considered feasible only if it increases the diversity of EC_v while preserving the l-diversity of both EC_v and EC_s. When multiple feasible swaps exist, the algorithm selects the swap that minimizes the increase in information loss. The process is repeated until EC_v satisfies l-diversity or no feasible swap can be identified. By repairing violations through record swaps, this phase preserves the existing equivalence class structure generated by the k-member algorithm and limits additional information loss.

After the swapping phase, unresolved violating ECs are

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Table 1 Attributes Used in the Adult Dataset

#	Attribute	Cardinality	Generalization (Height)
1	age	73	Taxonomy Tree (6)
2	workclass	9	Taxonomy Tree (3)
3	marital-status	7	Taxonomy Tree (2)
4	occupation	15	Taxonomy Tree (2)
5	relationship	6	Taxonomy Tree (2)
6	race	5	Taxonomy Tree (2)
7	sex	2	Taxonomy Tree (1)
8	native-country	42	Taxonomy Tree (3)
9	education	15	Taxonomy Tree (3)
10	income	2	Taxonomy Tree (2)

collected and processed in the rebuild phase. The records belonging to these unresolved ECs are pooled and reclustered using a k-member algorithm, producing a new equivalence classes compositions that may provide a more favorable distribution of sensitive values and re-evaluated for l-diversity. Finally, any ECs that still violate l-diversity are handled in the merge phase. For each unresolved EC, the algorithm identifies candidate merge partners and selects the equivalence class that results in the smallest increase in information loss after generalization. The selected ECs are then merged and updated accordingly.

3. Experimental Evaluation

3.1 Datasets

We used the *Adult* dataset [7], derived from the 1994 U.S. Census and publicly available through the UCI Machine Learning Repository. The Adult dataset is one of the most widely used benchmarks in privacy-preserving data publishing and anonymization research, enabling the evaluation of both privacy protection and data utility.

Table 1 summarizes the attributes used in this study. Eight attributes were selected as quasi-identifiers (QIDs): *age*, *workclass*, *marital-status*, *occupation*, *relationship*, *race*, *sex*, and *native-country*. These attributes were selected due to their high potential for re-identification when combined with external information [8]. The *education* attribute was selected as the sensitive attribute (SA), while the *income* attribute was used as the target class for machine learning classification.

3.2 Evaluation Metrics

Anonymization quality was evaluated across three dimensions: information loss metrics, impact on machine learning utility, and computational efficiency.

3.2.1 Normalized Certainty Penalty (NCP)

The working principle of NCP is to characterize the information loss of an entire anonymized partition by aggregating the penalties incurred by individual equivalence classes.

Let EC denote an equivalence class and let QI be a quasi-identifier attribute contained in EC. The definition of NCP depends on the attribute type. For a numerical quasi-identifier attribute QI^{Num}, the NCP of an equivalence class EC is defined as:

$$\mathrm{NCP}_{Num}(EC) = \frac{\max QI^{Num}(ec) - \min QI^{Num}(ec)}{\max QI^{Num} - \min QI^{Num}},$$

where the numerator represents the value range of Q^{Num} within the equivalence class T, and the denominator corresponds to the global range of the same attribute across the entire dataset.

For categorical quasi-identifier attributes QI^{Cat}, NCP is defined with respect to the taxonomy tree associated with the attribute. Let u denote the lowest common ancestor (LCA) of all values of QI^{Cat} appearing in the equivalence class EC. The categorical NCP is defined as:

$$\mathrm{NCP}_{QI^{Cat}}(EC) = \{\, 0 \,, if \mathrm{card}(u) = 1, \frac{\mathrm{card}(u)}{|QI^{Cat}|}, otherwise,$$

where $\mathrm{card}(u)$ denotes the number of leaf nodes in the subtree rooted at u, and $|QI^{Cat}|$ is the total number of distinct categorical values of the attribute.

Let d denote the total number of quasi-identifier attributes in the dataset, and let QI_i represent the i-th quasi-identifier attribute, which may be either numerical or categorical. Assuming a set of attribute weights $\{w_i\}_{i=1}^{d}$ such that $\sum_{i=1}^{d} w_i = 1$, the overall NCP of an equivalence class EC is computed as:

$$\mathrm{NCP}(EC) = \sum_{i=1}^{d} w_i \cdot \mathrm{NCP}_{Q_i}(EC).$$

3.2.2 Discernibility Metric (DM)

DM [9] assigns quadratic penalties to class sizes to quantify individual record indistinguishability:

$$\mathrm{DM} = \sum_{EC_i \in EC_{total}} |EC_i|^2 + \sum_{r \in Suppressed} |D|. \qquad (5)$$

where $|D|$ denotes the number of records in the original dataset and $|EC|$ denotes the size of an equivalence class. A lower DM value indicates lower information loss and therefore higher data utility, whereas larger values imply that records have been grouped into larger equivalence classes or suppressed to a greater extent.

3.2.3 Average Equivalence Class Size (CAVG)

This metric quantifies the extent of data loss introduced by the anonymization process by analyzing the average size of the generated equivalence classes [10]. It calculates the ratio between the total number of records and the product of num_eq and k.

$$\mathrm{CAVG} = \frac{|D|}{num_eq \times k}$$

where $|D|$ denotes the total number of records in the dataset, num_eq is the total number of equivalence classes, and k represents the privacy parameter. A higher value for CAVG suggests a more uniform distribution across equivalence classes, resulting in better utility and lower information loss.

This metric measures the degree of similarity between

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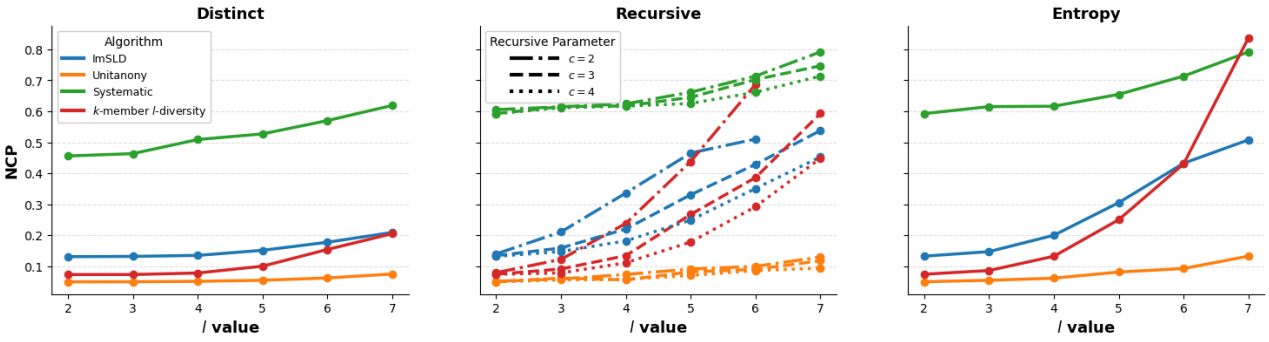


Fig. 1 Comparison of NCP under Distinct, Recursive, and Entropy l-diversity

the equivalence classes produced by the anonymization algorithm and the ideal configuration, in which each equivalence class contains exactly k records. Lower CAVG values approaching 1.0 indicate optimal partitions without over-generalization.

3.2.4 Predictive Utility (Classification Accuracy)

To evaluate predictive performance, a Random Forest classifier is trained and tested on both the original and anonymized datasets. Random Forest was selected due to its robustness and strong performance on datasets containing both numerical and categorical attributes. Classification accuracy is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN},$$

where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively. Higher accuracy values indicate that the anonymized data retains more predictive information from the original dataset, while lower values indicate a greater loss of utility. The classification accuracies obtained from the original and anonymized datasets are compared to assess the impact of anonymization on data utility.

3.2.5 Disclosure Risk Metrics

Average Disclosure Risk (ADR) and Worst-Case Disclosure Risk (WDR) is used to evaluate the resistance of anonymized datasets against sensitive attribute inference attacks. These metrics estimate the probability that an attacker correctly infers a sensitive attribute by selecting the most frequent sensitive value within an equivalence class.

Let EC_j denote an equivalence class containing n_j records, N denote the total number of records in the dataset, and $P(S = s|EC_j)$ denote the probability of sensitive value s occurring in EC_j.

The disclosure risk of an equivalence class is defined as

$$Risk(EC_j) = \max_{s \in S} P(S = s \mid EC_j). \tag{6}$$

Average Disclosure Risk measures the expected probability that an attacker correctly infers the sensitive attribute across the entire dataset. It is computed as

$$ADR = \sum_{j=1}^{m} \frac{n_j}{N} Risk(EC_j), \tag{7}$$

where m is the number of equivalence classes. A lower ADR indicates a lower average probability of successful attribute inference and therefore stronger privacy protection.

Worst-Case Disclosure Risk measures the maximum disclosure risk among all equivalence classes. It is defined as

$$WDR = \max_{j} \left(Risk(EC_j) \right), \tag{8}$$

WDR represents the highest attacker success probability in the dataset. A lower WDR indicates that no equivalence class exposes a highly predictable sensitive value and therefore provides stronger protection against attribute inference attacks.

3.3 Experimental Setup

Experiments were executed on an HPE Cray XD2000 cluster node equipped with AMD EPYC 7763 CPUs and 740 GB RAM. The parameter k was fixed at 10, while l ranged from 2 to 7. For Recursive (c, l)-diversity, c was set from 2 to 4.

3.4 Result

3.4.1 NCP Analysis

Figure 1 shows the NCP values across all privacy models. Across all configurations, NCP exhibits a monotonic increase with respect to l, reflecting the necessity of broader QID generalizations to satisfy tighter sensitive attribute diversity constraints.

Distinct l-diversity produces the lowest NCP values, suggesting that it preserves data utility effectively. UnitAnony and the k-member repair algorithm achieve the lowest penalty profiles, maintaining NCP values below 0.08 and 0.21, respectively, even at $l = 7$.

Recursive (c, l)-diversity results in higher information loss than Distinct l-diversity due to the additional constraints imposed on sensitive value distributions. Lower values of the recursive parameter ($c = 2$) enforce stricter diversity constraints, resulting in a rapid increase in NCP for $l \geq 4$. At $l = 7$, both ImSLD and the k-member repair algorithm reach an NCP value of 1, indicating complete information loss caused by excessive generalization.

Entropy l-diversity produces the highest NCP values, particularly at larger l values, due to the need to maintain a uniform spread of sensitive attributes within every equivalence class.

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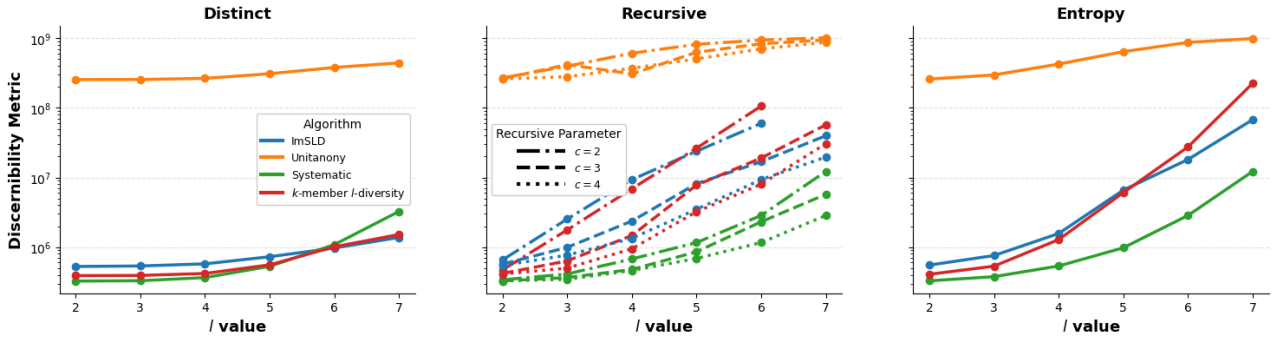


Fig. 2 Comparison of Discernibility Metric (DM) under Distinct, Recursive, and Entropy l-diversity.

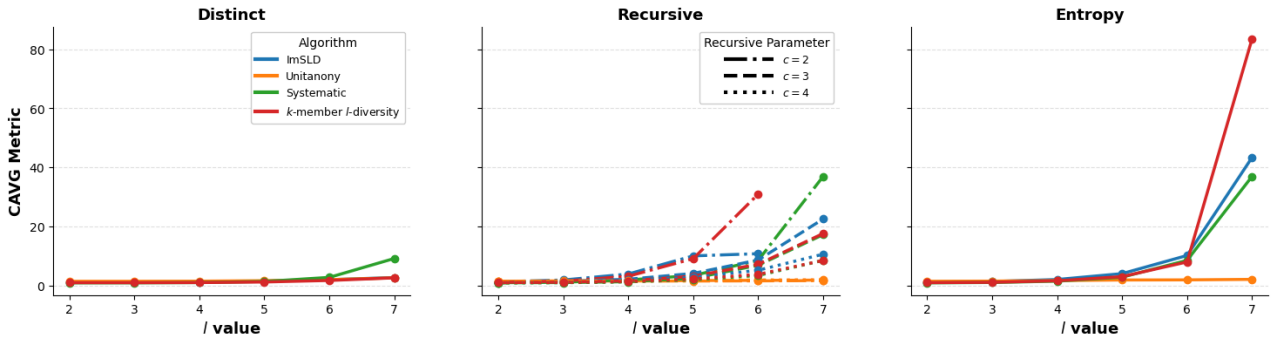


Fig. 3 Comparison of CAVG values under Distinct, Recursive, and Entropy l-diversity.

3.4.2 Discernibility analysis

Figure 2 presents the DM values for the evaluated algorithms. Similar to NCP, DM increases as the privacy requirements become stronger. UnitAnony yields a significantly higher level of discernibility compared to all algorithms across all three diversity models due to record suppression when a candidate lattice scheme fails to satisfy the l-diversity requirement.

In the Recursive and Entropy configurations, the k-member repair algorithm achieves performance comparable to ImSLD at low to moderate diversity levels ($l \leq 5$). However, under the Entropy l-diversity configuration with $l = 7$, the DM value of the proposed method increases sharply. This observation further supports the NCP results, indicating that the repairing process leads to substantially larger equivalence classes (ECs).

3.4.3 Equivalence class analysis

The structural granularity of the anonymized partitions is detailed in Figure 3 via the CAVG metric. Distinct l-diversity exhibits highly stable class sizes across all algorithms, with CAVG values remaining between 1.1 and 3.2 up to $l = 6$. Under Recursive and Entropy models, CAVG behavior bifurcates. For Recursive diversity at $c = 2$ and $l = 7$, the k-member repair algorithm reaches a CAVG of 37.2. Under Entropy l-diversity at $l = 7$, the k-member repairing algorithm spikes abruptly to 83.4. This behavior confirms that enforcing high entropy distributions on highly skewed empirical datasets forces the repair mechanism to aggregate numerous smaller clusters into large during merging equivalence class.

3.4.4 Prediction performance analysis

Figure 4 presents the decreasing machine learning utility of anonymized datasets. In general, predictive performance decreases as privacy requirements become stronger. The horizontal dashed line denotes the baseline classification accuracy (80.3%) achieved on the original dataset.

Distinct l-diversity preserves classification accuracy most effectively, with only minor changes as l increases. Recursive l-diversity introduces a moderate reduction in accuracy due to the additional privacy constraints. Entropy l-diversity causes the largest decrease in predictive performance, particularly at higher values of l, where extensive data modifications are required.

3.4.5 Disclosure risk analysis

Figure 5 and 6 presents the disclosure risk, which measures the average disclosure risk (ADR) and worst-case disclosure risk (WDR) probability that an attacker correctly infers a sensitive value. Unlike the utility metrics, lower disclosure risk values indicate better privacy protection.

Distinct l-diversity provides the weakest protection against attribute disclosure, with ADR values declining marginally from 0.46 at $l = 2$ to 0.41 at $l = 7$ for the k-member repair algorithm, ImSLD, and UnitAnony algorithms. Recursive (c, l)-diversity provides substantially enhanced protection. Decreasing c from 4 to 2 reduces ADR from 0.35 down to 0.26 at $l = 7$, successfully addressing the probability of dominant attribute inference. Entropy l-diversity achieves the lowest ADR values, providing the strongest resistance to attribute inference attacks.

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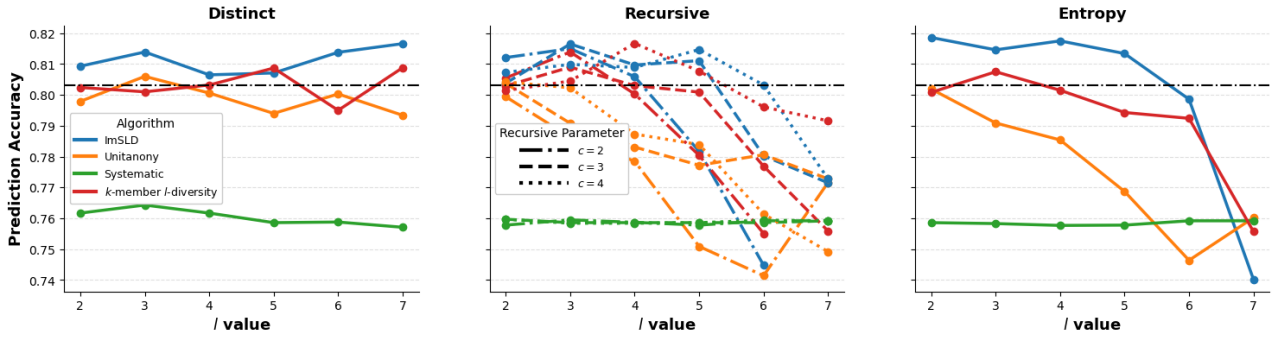


Fig. 4 Comparison of classification accuracy under Distinct, Recursive, and Entropy l-diversity.

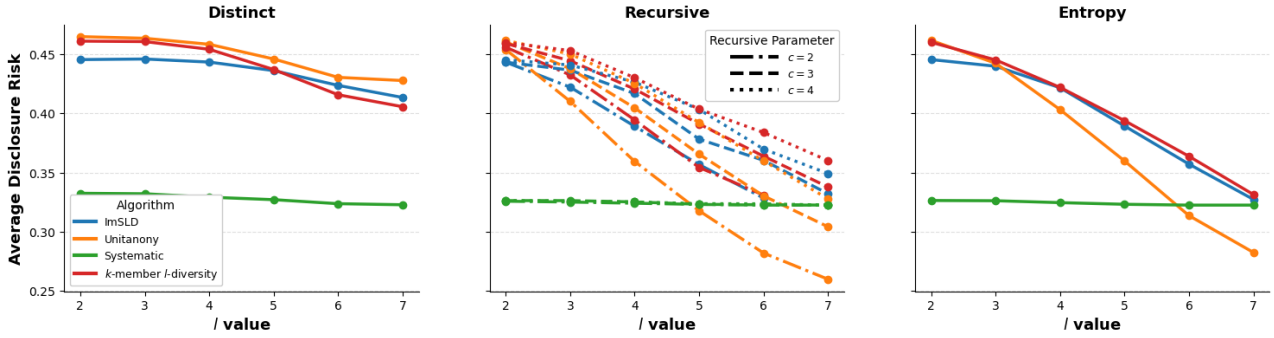


Fig. 5 Comparison of Average Disclosure Risk (ADR) under Distinct, Recursive, and Entropy l-diversity.

3.4.6 Execution time analysis

Figure 7 presents the execution time of the evaluated algorithms under Distinct, Recursive, and Entropy l-diversity. Execution time is a critical determinant of practical applicability, as privacy-preserving data publishing algorithms must maintain computational tractability when scaling to large-scale microdata workloads.

For Distinct l-diversity, ImSLD consistently achieves the best efficiency for execution time and is unaffected by increasing values of l. In contrast, UnitAnony and Systematic require substantially longer processing times. The k-member l-diversity algorithm requires more computation than ImSLD and remains comparable to the other repair-based approaches.

Under Recursive l-diversity, execution time generally increases across all algorithms due to additional checks that are required to enforce the recursive diversity constraint. The results show that the k-member l-diversity algorithm maintains execution times comparable to Systematic and UnitAnony while providing stronger privacy guarantees. ImSLD remains the most computationally efficient algorithm across all parameter settings.

For Entropy l-diversity, execution times are similar to those observed for Distinct l-diversity. ImSLD continues to exhibit the lowest execution time, whereas UnitAnony and Systematic remain the most computationally expensive methods. The k-member repair algorithm requires additional processing to satisfy entropy constraints but maintains stable execution times across different values of l.

The empirical findings highlight clear trade-offs across the evaluated privacy models and anonymization algorithms:

(1) **Distinct l-Diversity** is best suited for applications that prioritize high data utility, feature granularity, and classification accuracy, provided that the sensitive attribute distribution is relatively uniform.

(2) **Recursive (c, l)-Diversity** provides the most balanced trade-off between privacy and utility. Adjusting the parameter c enables data publishers to control the dominance threshold, thereby reducing disclosure risk while avoiding excessive information loss. Among the evaluated configurations, $c = 3$ consistently achieved the best overall performance across all evaluation metrics.

(3) **Entropy l-Diversity** is most appropriate for high-security applications where minimizing attribute disclosure is the primary objective. However, its stringent diversity constraints lead to substantially larger equivalence classes and significant utility degradation, particularly when $l \geq 6$.

(4) **Algorithmic selection** The k-member repair algorithm consistently achieves superior utility preservation and compact equivalence classes under Distinct and Recursive l-diversity ($l \leq 5$). UnitAnony provides a practical alternative when minimal quasi-identifier distortion is required and record suppression is acceptable, whereas Systematic Clustering is limited by comparatively high baseline information loss.

(5) **Computational efficiency.** ImSLD consistently

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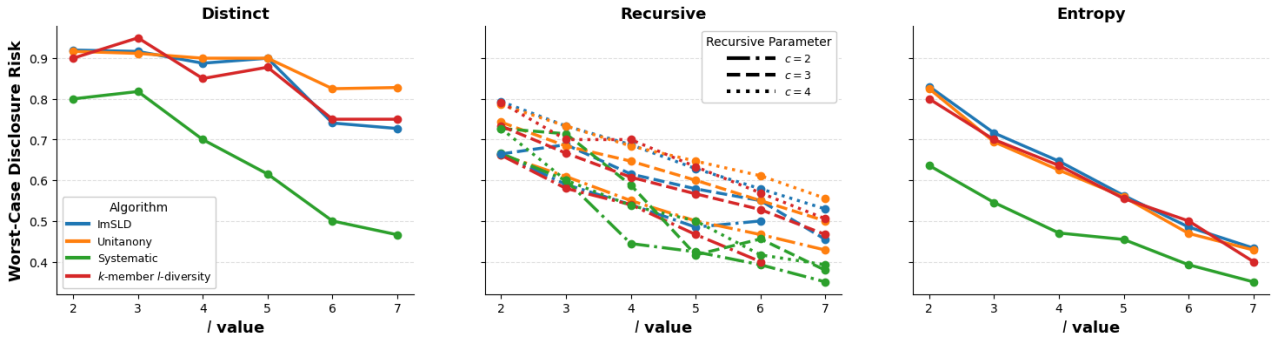


Fig. 6 Comparison of Worst-case Disclosure Risk (WDR) under Distinct, Recursive, and Entropy l-diversity.

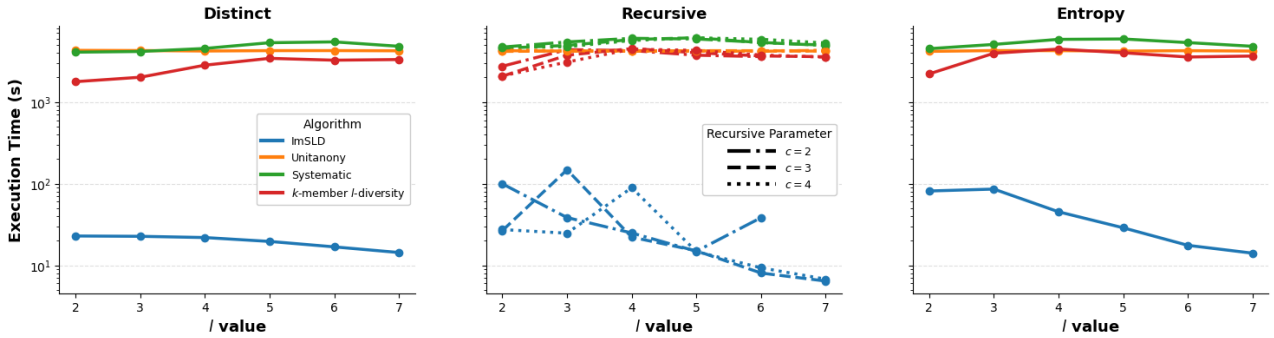


Fig. 7 Comparison of execution time under Distinct, Recursive, and Entropy l-diversity.

achieves the shortest execution time across all l-diversity models. While UnitAnony, Systematic Clustering, and the k-member repair algorithm require longer execution times, the computational overhead remains moderate. Therefore, privacy and utility considerations are more important than execution time when selecting an anonymization method.

4. Conclusion

This study compared Distinct, Recursive, and Entropy l-diversity using four anonymization algorithms: ImSLD, UnitAnony, Systematic Clustering, and the k-member repair algorithm. The results demonstrate a the trade-off between privacy protection and data utility. Distinct l-diversity preserves data utility most effectively but provides limited protection against sensitive attribute disclosure. In contrast, Entropy l-diversity offers the strongest privacy protection at the cost of higher information loss and lower classification accuracy. Recursive (c, l)-diversity achieves the most balanced trade-off between privacy and utility, with $c = 3$ providing the most favorable overall performance by significantly reducing disclosure risk while preserving data quality and classification accuracy

The k-member repair algorithm achieves competitive results across all evaluation metrics, particularly under low-to-moderate diversity requirements ($l \leq 5$), where it effectively preserves data utility while satisfying privacy constraints.

Future work will investigate more efficient repair strategies for Entropy l-diversity, evaluate the approach on high-dimensional datasets and multiple sensitive attributes, and explore the integration of differential privacy to further enhance protection against inference attacks.

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